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LinkedHashSet, SortedSet, NavigableSet in Java

=> LinkedHashSet :-

- ➔ LinkedHashSet is the child class of HashSet which is present in java.util package
- ➔ Syntax : public class LinkedHashSet extends HashSet implements Set, Cloneable, Serializable { - }
- ➔ The underline data structure of LinkedHashSet is "Hashtable + LinkedList"
- ➔ LinkedHashSet was introduced in JDK 1.4 version

➔ Properties of LinkedHashSet :-

= All properties of LinkedHashSet is same as HashSet except LinkedHashSet follows the insertion order

➔ Constructors of LinkedHashSet :-

= same constructors as HashSet

➔ Methods of LinkedHashSet :-

= Same methods as HashSet

➔ When we should use LinkedHashSet :-

= If we have to create cache based applications then we can use LinkedHashSet

➔ Difference between HashSet & LinkedHashSet :-

1. HashSet was introduced in JDK 1.2 version

LinkedHashSet was introduced in J2SE 1.4 version

2. HashSet does not follow the insertion order

LinkedHashSet follows the insertion order

3. HashSet underline data structure is "Hashtable"

LinkedHashSet underline data structure is "Hashtable + LinkedList"

=> SortedSet :-

- ➔ SortedSet is a child interface of Set interface which is present in java.util package
- ➔ Syntax : public interface SortedSet extends Set { - }
- ➔ SortedSet was introduced in JDK 1.2 version

➔ Properties of SortedSet :-

1. SortedSet is not an index based data structure
2. SortedSet does not follows the insertion order
3. SortedSet follows the sorting order
4. SortedSet can store same data types or homogeneous elements. If we provide different data type element if will provide exception i.e. java.lang.ClassCastException
5. SortedSet cannot store the duplicate elements
6. We should not store null value in SortedSet because SortedSet follows the sorting order so while comparing the elements with null value, it will provide the java.lang.NullPointerException
7. SortedSet allows Comparable objects bydefault, but if we insert non-comparable objects, then it will provide an exception i.e. java.lang.ClassCastException

8. SortedSet is non-synchronized collection

➔ Methods of SortedSet :-

1. Object first();
2. Object last();
3. SortedSet headSet(Object toElement);
4. SortedSet tailSet(Object fromElement);
5. SortedSet subSet(Object fromElement, Object toElement);

SortedSet headSet() Method can present element after elements are printed

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=> NavigableSet :-

➔ NavigableSet is the child interface of SortedSet interface which is present in java.util package

➔ Syntax : public interface NavigableSet extends SortedSet { - }

➔ NavigableSet was introduced in Java SE 6 version

➔ Properties of NavigableSet :-

= NavigableSet has the same properties as that of SortedSet but it provides some extra navigable methods

➔ Methods :

1. public NavigableSet descendingSet()
2. public Object ceiling(Object obj)
3. public Object higher(Object obj)
4. public Object floor(Object obj)
5. public Object lower(Object obj)
6. public Object pollFirst()
7. public Object pollLast()

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